[CSED233] Data Structure

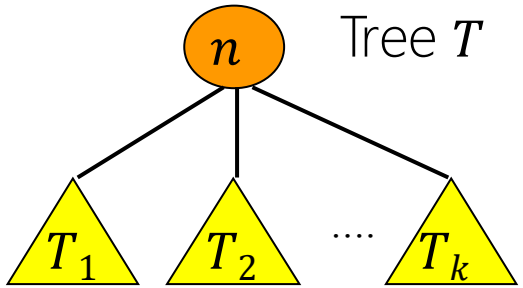
Tree (2)

Seung-Hwan Baek

POSTECH

Recap: Tree Traversal

- Types of traversals

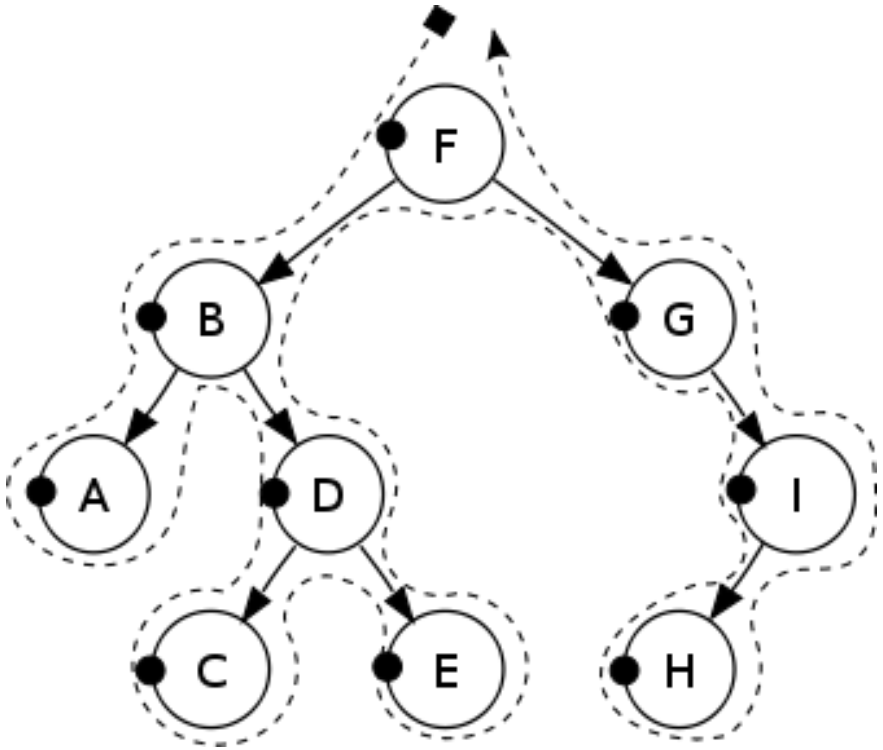


- $Preorder(T) = < n,\ Preorder(T_1), \dots, Preorder(T_k) >$
- $Postorder(T) = < Postorder(T_1), \dots, Postorder(T_k),\ n >$
- $Inorder(T) = < Inorder(T_1),\ n,\ Inorder(T_2), \dots, Inorder(T_k) >$

Binary Tree Traversals

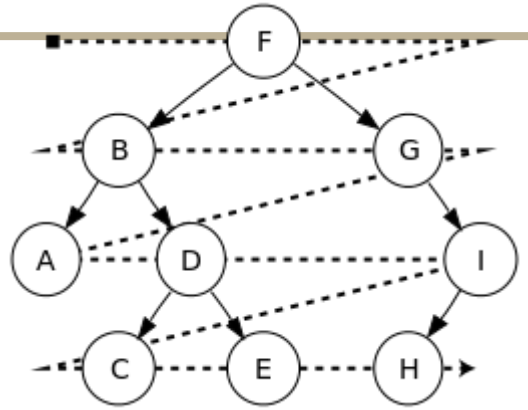
- Depth-first search
  - Go down first
  - Recursive way – via (call) STACK
  - 3 variations: pre-order, post-order, in-order

```
While visiting all nodes {
  current node = stack.pop()
  For each child of the current node
      stack.push( child )
}
```



- Time complexity: $O(n)$, where n is # of nodes in a tree

Binary Tree Traversals



- Breadth-first search
  - Go across first (same level)
  - Implemented via QUEUE
  - Level-order

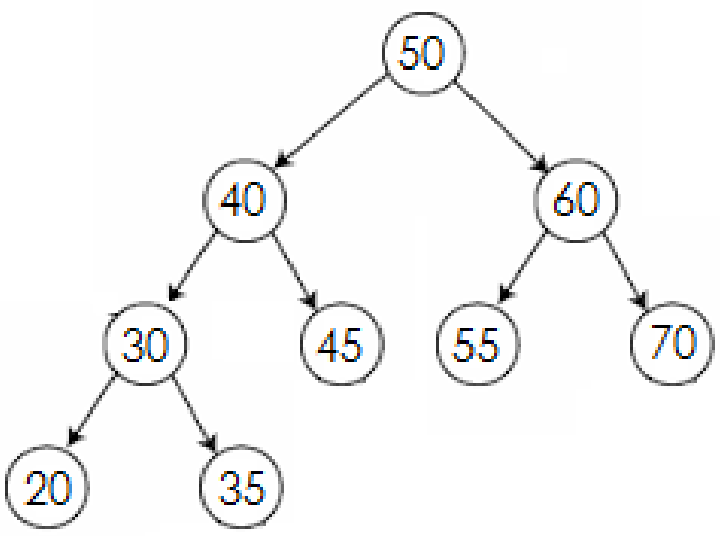
```
While visiting all nodes {
  current node = queue.dequeue()
  For the current node's child
      queue.enqueue( child )
}
```

- Time complexity: $O(n)$, where n is # of nodes in a tree

Quiz

- We don't know the exact structure of a binary tree,
- If we know the traversals of the binary tree, can we reconstruct the true structure of the tree?

Binary Tree



Traversals
- Inorder
- Postorder
- Preorder

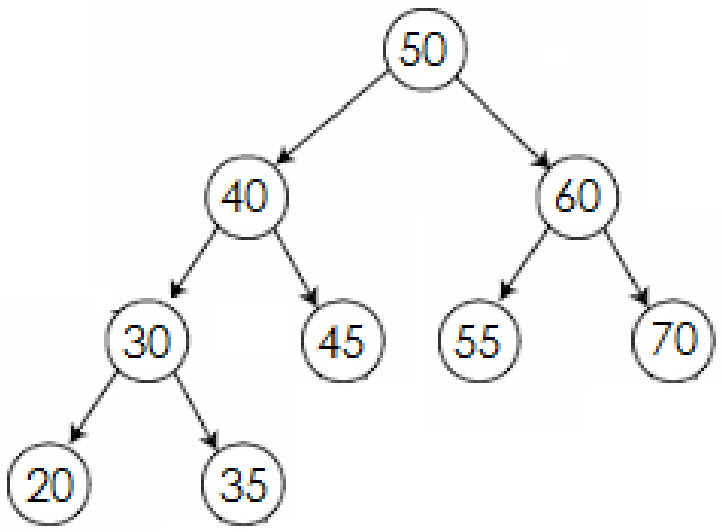
Single Traversal

- Any single traversal sequence (pre-, in- or post-order) cannot describes the underlying tree uniquely

Binary Tree

- $Inorder \ = < 20,30,35,40,45,50,55,60,70 >$

- $Postorder \ = < 20,35,30,45,40,55,70,60,50 >$



Unique Binary Tree (by Two Traversals)

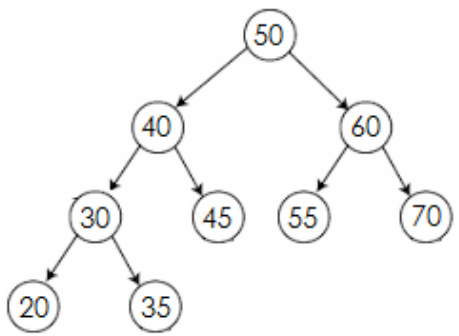
- We can identify the binary tree uniquely by two traversal sequences like:
  - (postorder & inorder), (preorder & inorder), (level-order & inorder)
    - inorder: to find Left & Right child/subtrees
    - postorder: to find the Root (the last in postorder)
    - preorder: to find the Root (the first/ in preorder)
    - level-order: to find the Root

- The other combinations leaves some ambiguity in the tree structure

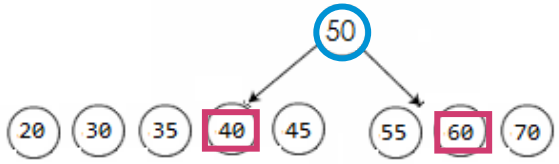
Binary Tree (by Two Traversals)

- Example:

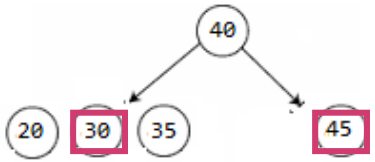
Binary Tree

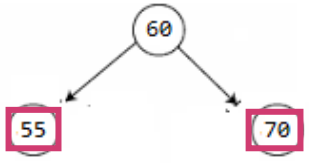



```
int inOrder[]   = {20, 30, 35, 40, 45, 50, 55, 60, 70};
int postOrder[] = {20, 35, 30, 45, 40, 55, 70, 60, 50};
```

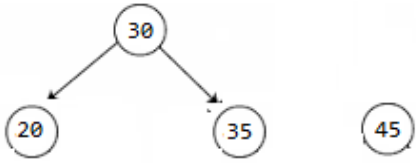



```
int inOrder[]   = {20, 30, 35, 40, 45,  ✗, 55, 60, 70};
int postOrder[] = {20, 35, 30, 45, 40}  55, 70, 60, ✗};
```






```
int inOrder[] =   {20, 30, 35,  ✗, 45  ✗, 55  ✗ 70};
int postOrder[] = {20, 35, 30   45 ✗,  55  70 ✗ ✗};
```



```
int inOrder[]   = 20   ✗, 35
int postOrder[] = 20   35 ✗,
```

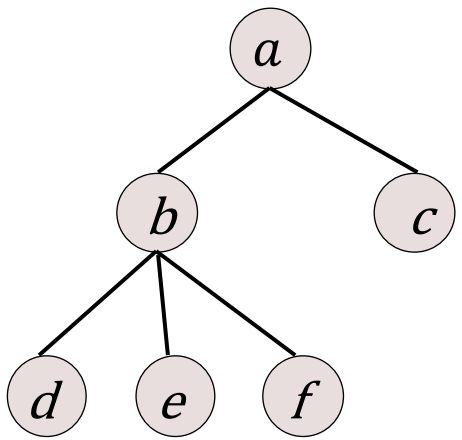
Reference: https://javabypatel.blogspot.com/2016/05/construct-binary-tree-from-inorder-and-post-order-traversals.html

Algorithm

- Algorithm (when Preorder & Inorder is given)
  - (1) $Root$ = the first node in the preorder
  - (2) Find the left child:
    - $Left_subtree$ = all nodes that are left to the $Root$ in the inorder
    - $Left_child$ = the first node in the preorder of the $Left_subtree$
  - (3) In the same way, find the right child:
    - $Right_subtree$ = all nodes that are right to the $Root$ in the inorder
    - $Right_child$ = the first node in the preorder of the $Right_subtree$
  - (4) Repeat the steps 2 & 3 with each new node (until every node is not visited in preorder)

General Tree Implementations

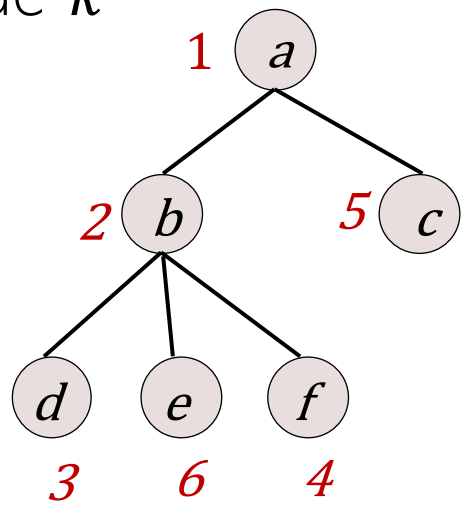
- General k-ary tree is a tree in which each node has no more than k children
- Implementations
  - Simple array implementation
  - List-of-Children implementation
  - Left-Child/Right-Sibling implementation
  - …



- Again, there would be other implementations as well
- No rules here, they have their own pros/cons
- Let's analyze them now

Simple Approach

- Parent point representation
- Two linear arrays – parallel to each other
  - $Tree[k] = m \Leftrightarrow$ node m is the parent of node k
  - $Label[k]$: the label (value, info) of node k



- Of course, you can also use structure

	1	2	3	4	5	6	…	n
$Tree[]$	0	1	2	2	1	2		
$Label[]$	a	b	d	f	c	e		

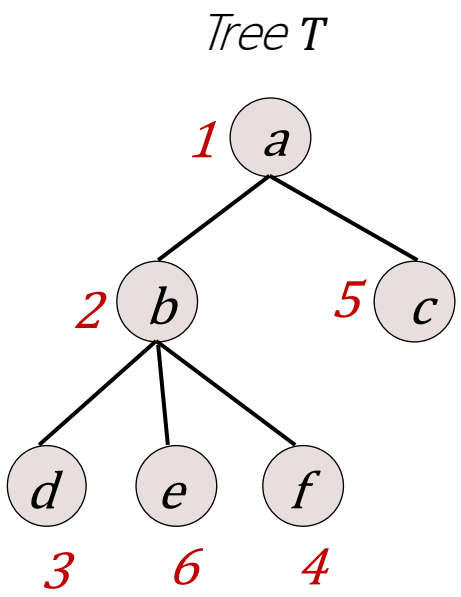
Not specify the order of children

Simple Approach: Time Complexity

- $Parent(k, T)$
  - What is the parent node of the node index k in the tree T?
  - Easy to compute: $O(1)$

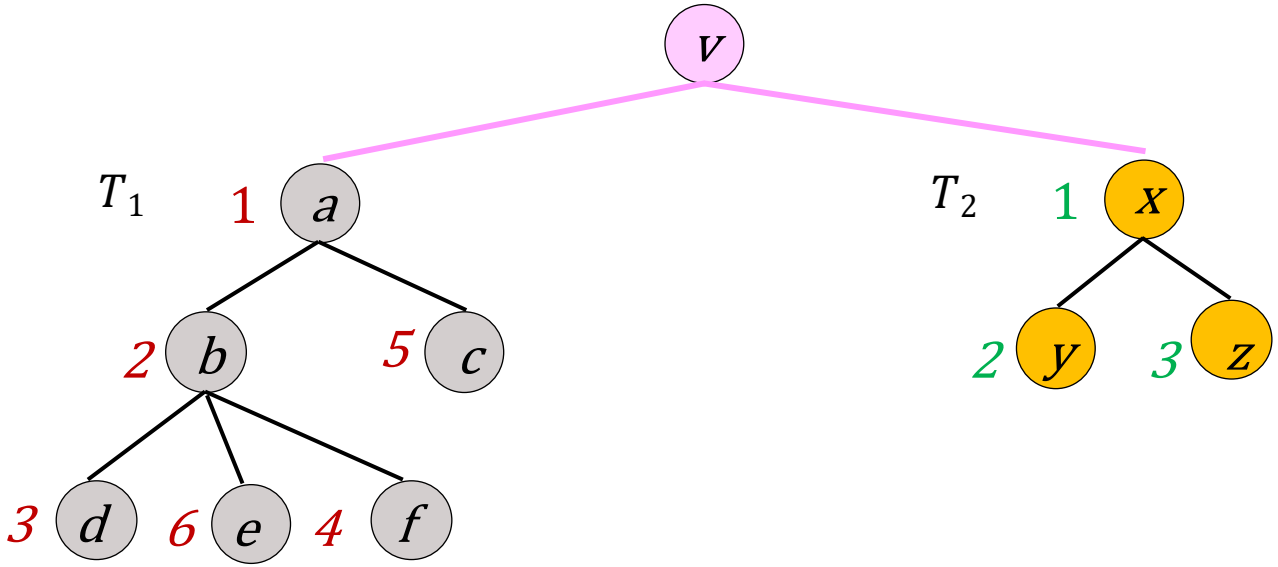
- $Leftmost_Child(k, T)$
  - What is the leftmost child of the node index k in the tree T?
  - Not well defined (no order of children)
  - $O(n)$

	1	2	3	4	5	6	…	n
$Tree[]$	0	1	2	2	1	2		
$Label[]$	a	b	d	f	c	e		



Simple Approach: Time Complexity

- $Create_m(v, T_1, T_2, \dots, T_m)$
  - Make a tree with a root v and subtrees $T_1, T_2, \dots, T_m$

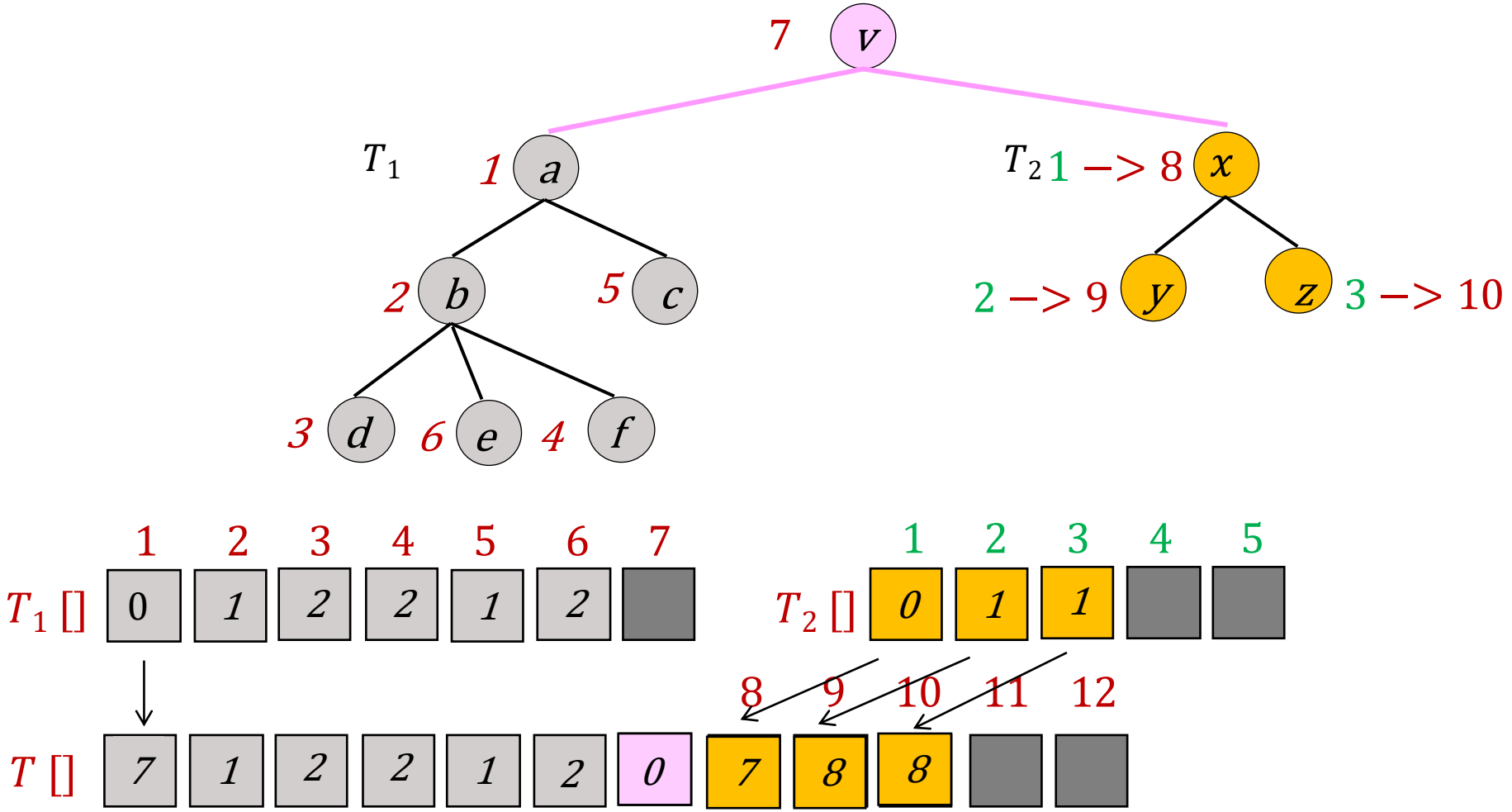


	1	2	3	4	5	6	…	n
T_1 []	0	1	2	2	1	2		
$Label$[]	a	b	d	f	c	e		

	1	2	3	4	5
T_2 []	0	1	1		

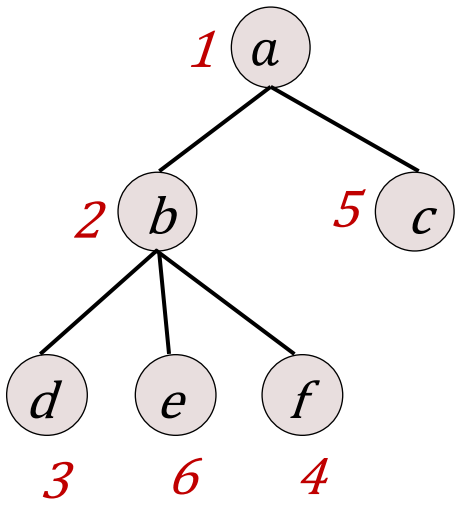
Simple Approach: Time Complexity

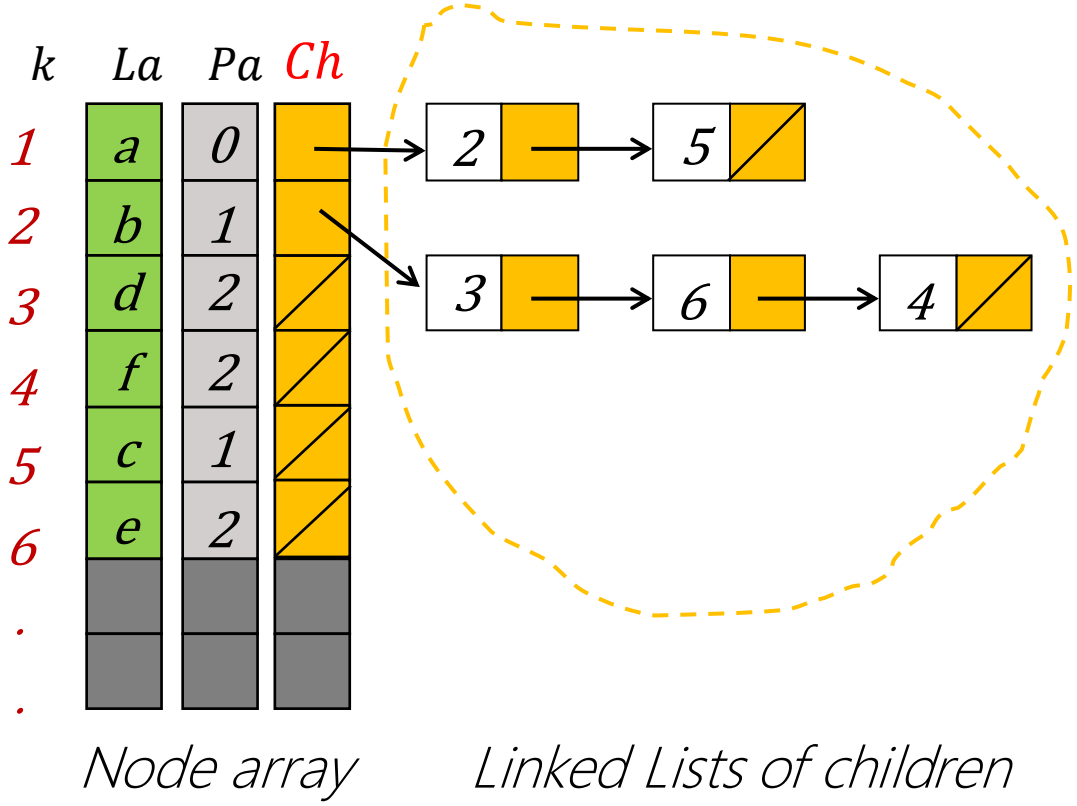
- $Create_m(v, T_1, T_2, \dots, T_m)$
  - Quite hard to handle as we should change the tree indices for $T_2, \dots, T_m$



List-of-Children Approach

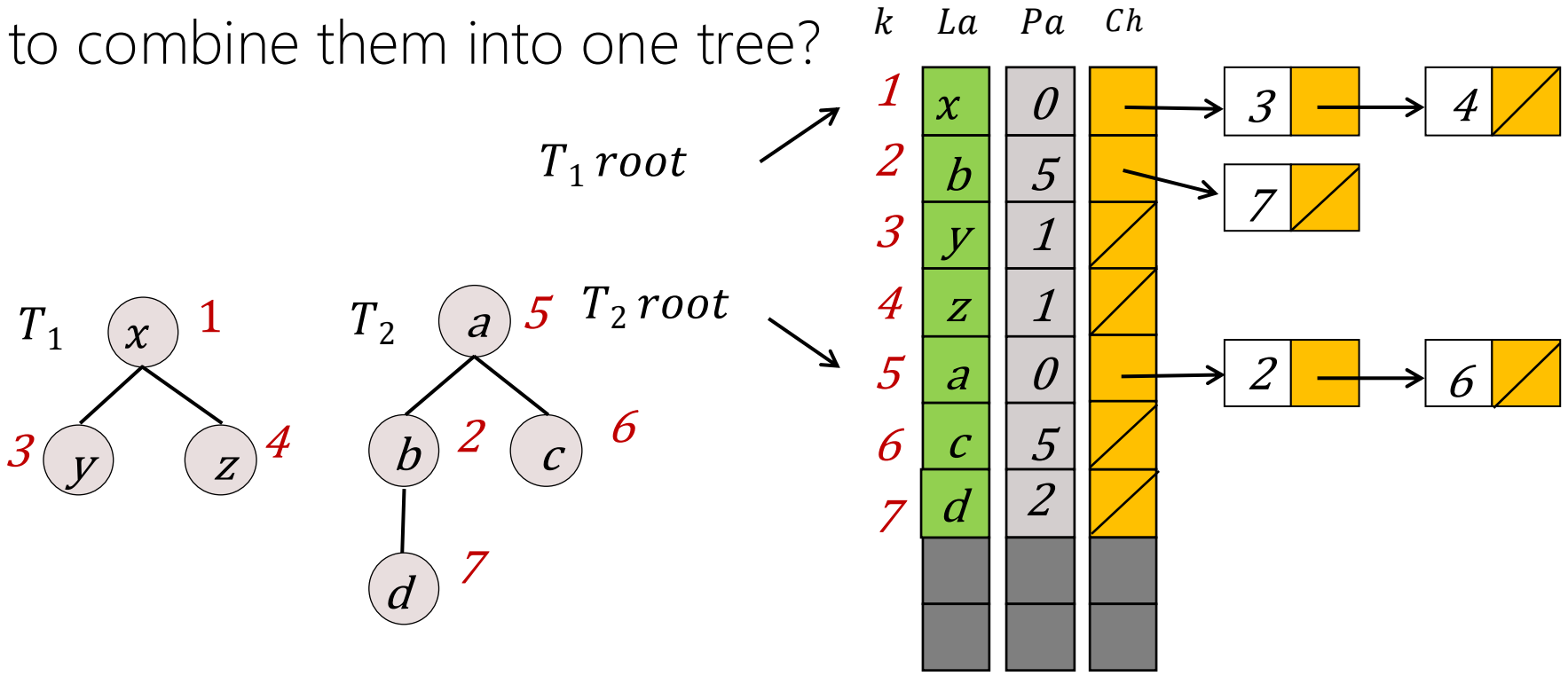
- Similar to the simple approach, we have the tree and the value arrays
- But, we have an additional array
  - Linked list of children (for each non-terminal node: (a, b) in this example)
  - Idea: let's keep both parent and children information!
- Node array (for all tree nodes)





List-of-Children Approach: Combining Trees

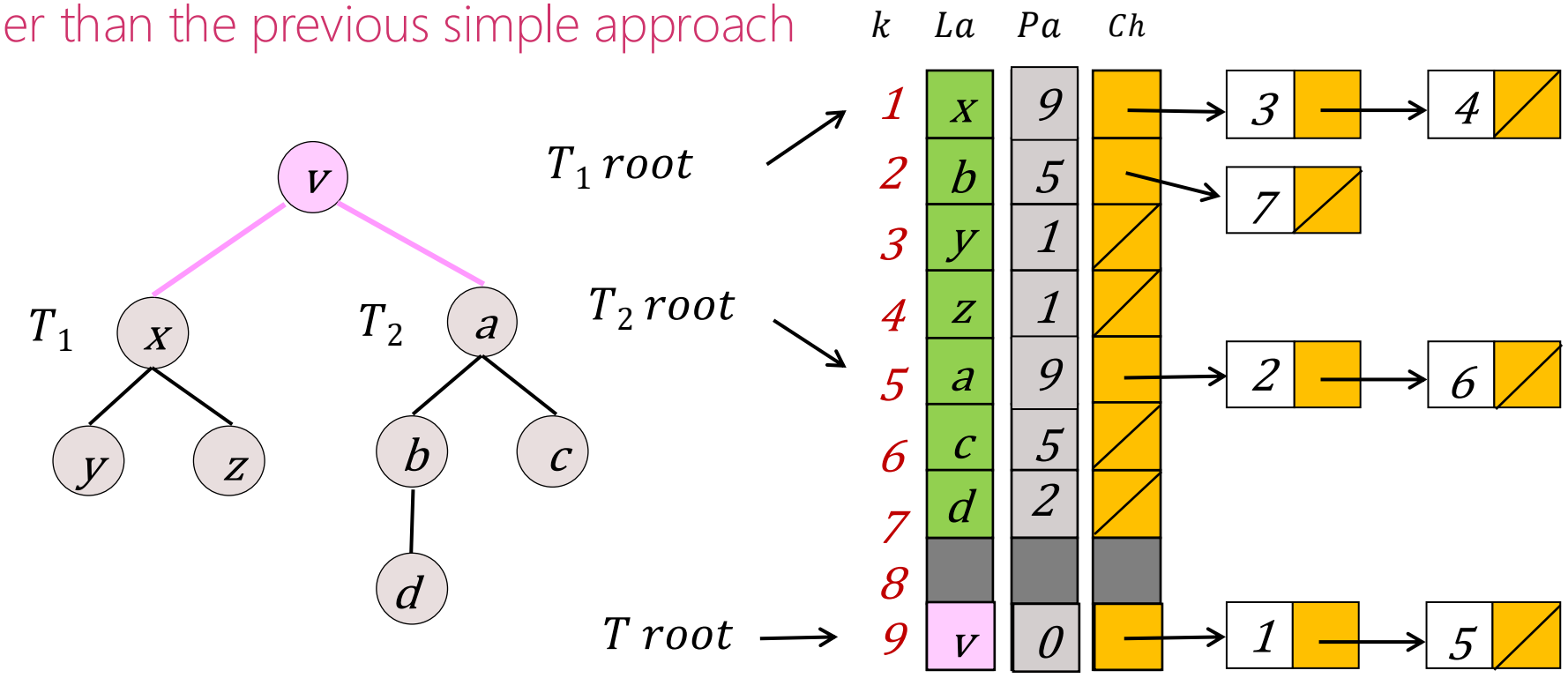
- $Create_2(v, T_1, T_2)$
- If we saved two trees T_1 and T_2 *in a single array*
- How to combine them into one tree?



Node array (for all trees)

List-of-Children Approach: Combining Trees

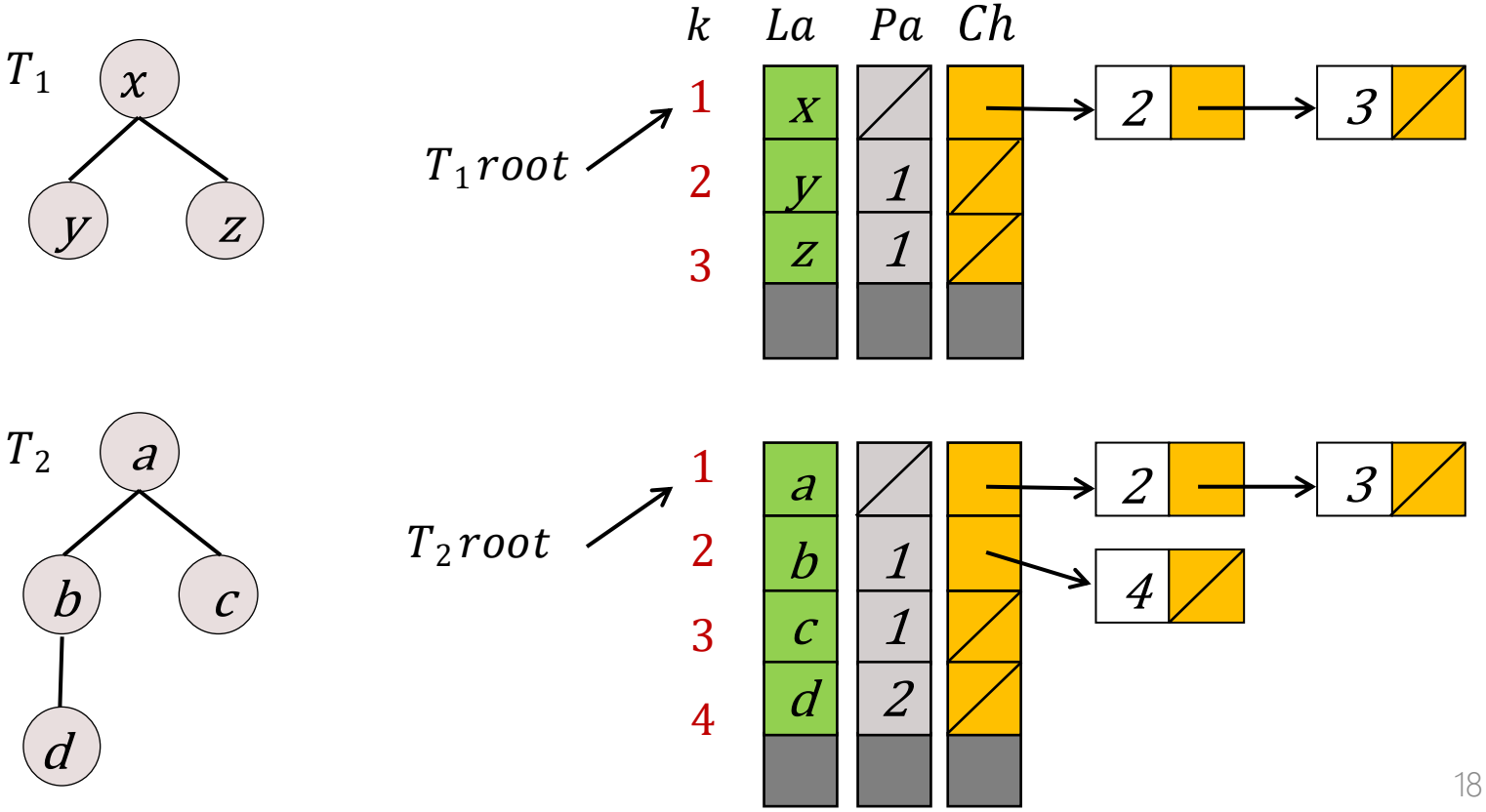
- $Create_2(v, T_1, T_2)$
  - Combining trees takes time complexity $O(1)$ (in case of binary tree)
    - Change the parents of the subtrees' roots
    - Easier than the previous simple approach



Node array (for all trees)

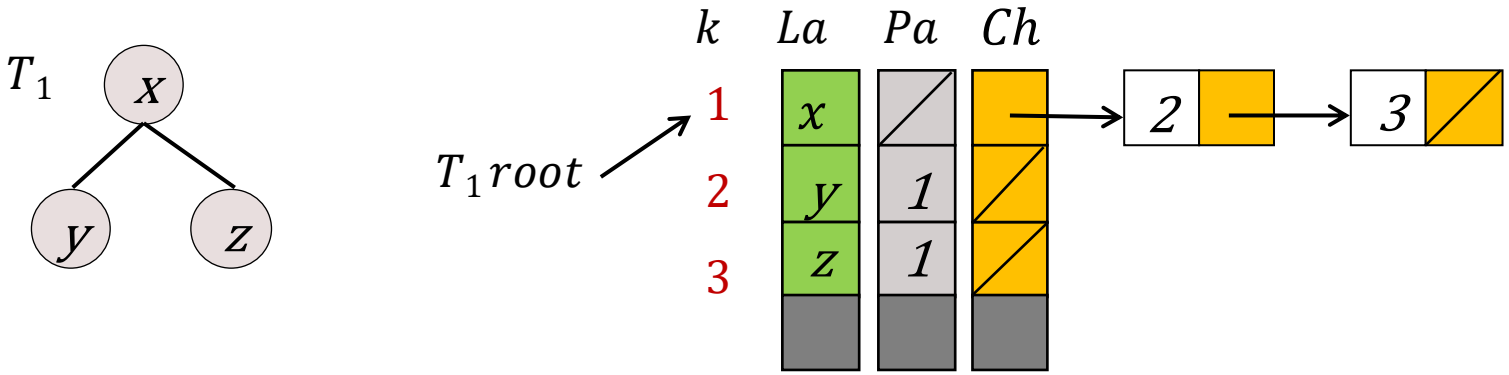
List-of-Children Approach: Combining Trees

- However… If each tree is stored in a separate array
  - Combining trees is OK?: *more cumbersome*
  - If we want to make it as a single array after the operation, again the second subtree's indices should be changed, resulting $O(n)$…



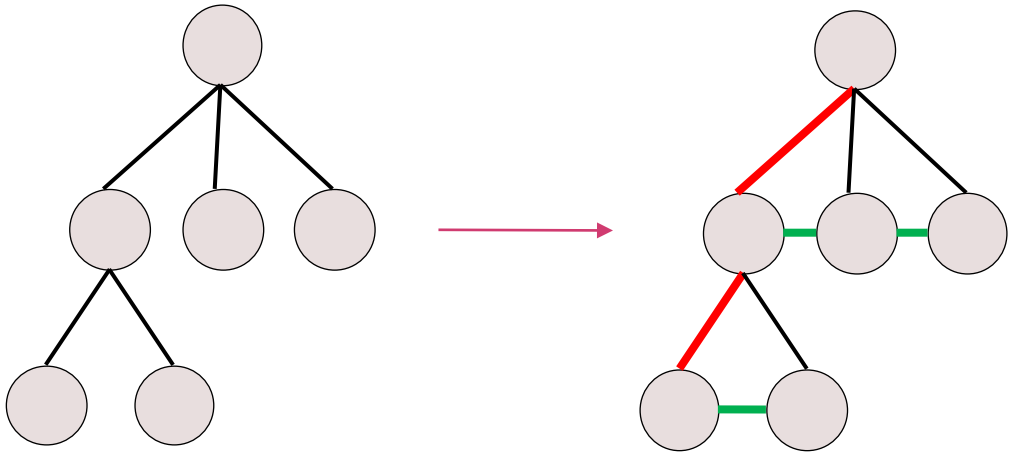
List-of-Children Approach: Problems

- Additional problems
  - Difficult to access a node's right sibling
    - How to access y's sibling?
      - Access parent's children list
  - Duplication of nodes
    - Each node appears both in the node array & in the linked list of children
      - Node 2's information is in the original list as well as the children list of its parent



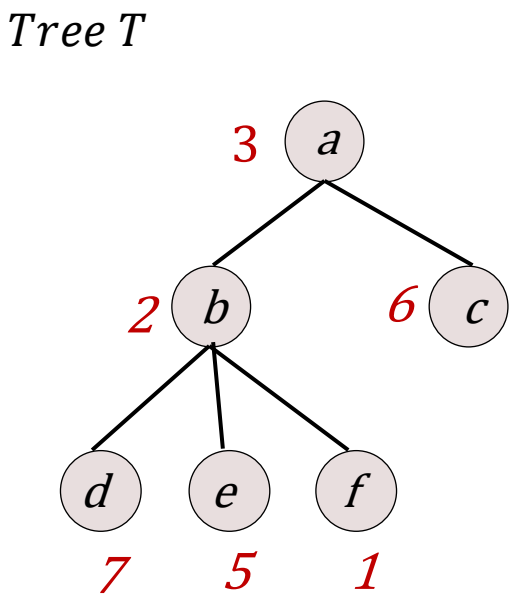
Left-Child/Right-Sibling Approach

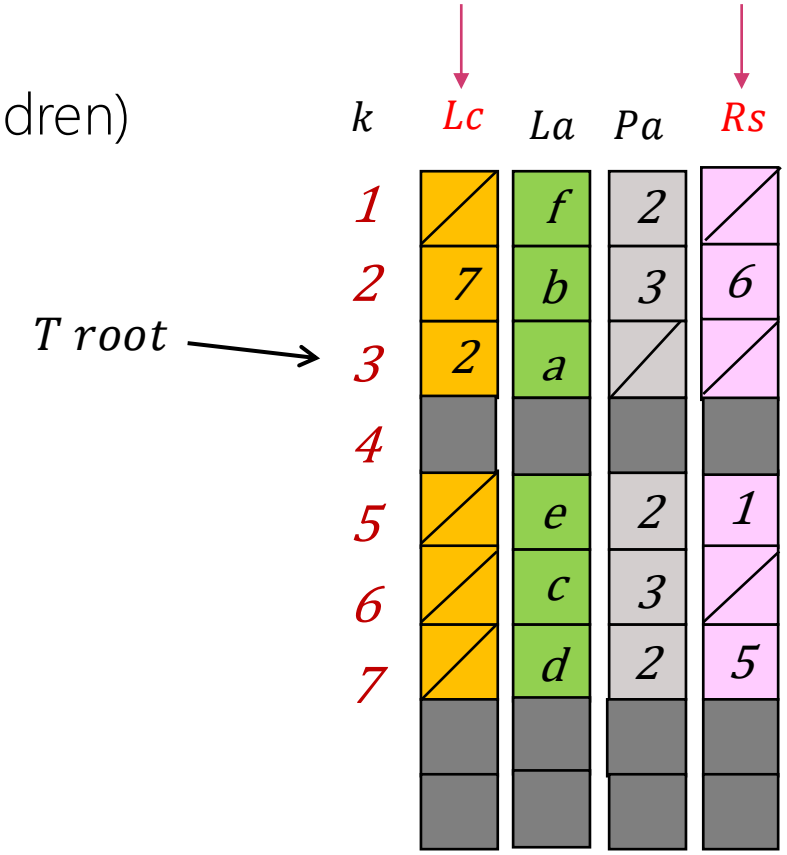
- Save left child and right sibling for each node



Left-Child/Right-Sibling Approach: Static

- Static implementation (using array)
  - Each node has pointers to its *parent*, *leftmost child*, & *right sibling*
  - Efficient for any # of children
    - (we don't need to store all the children)

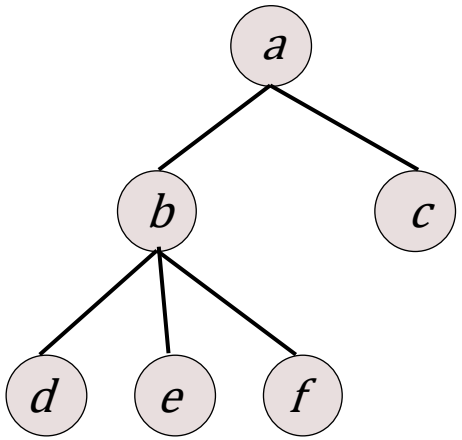


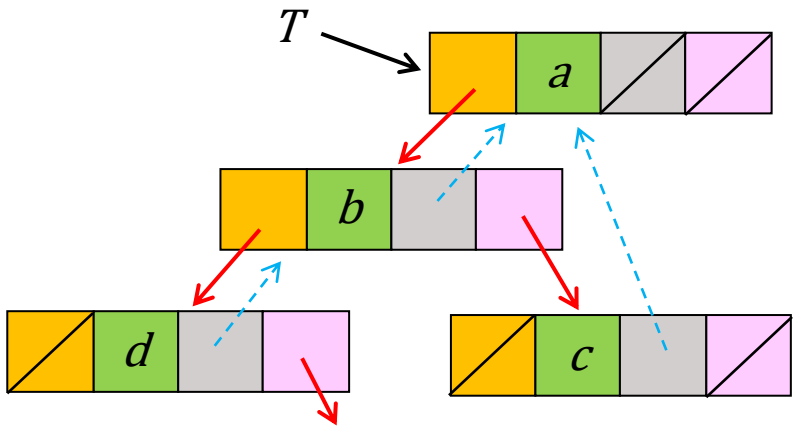


Left-Child/Right-Sibling Approach: Dynamic

- Problem of static *left-child/right-sibling* approach
  - Using an array to store the collection of nodes
  - Fixed space for nodes (no dynamic space allocation)

- Can we use linked implementation – dynamic one?

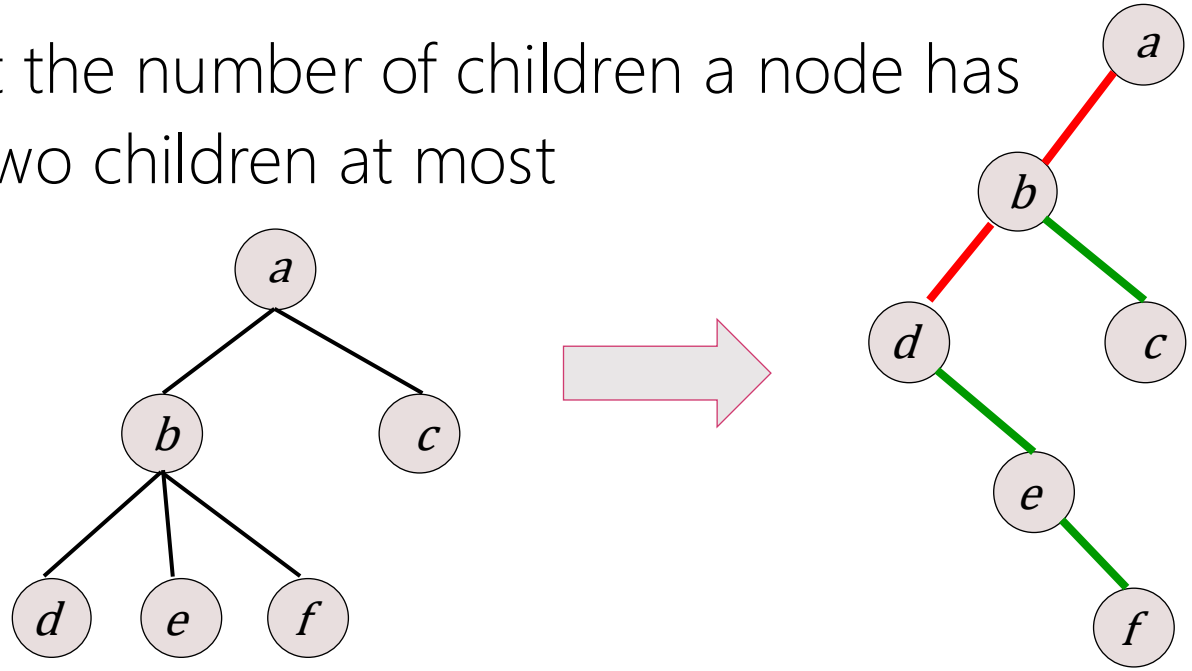




Left-Child/Right-Sibling Approach

- k-ary tree is a tree in which each node has no more than k children

- Any general k-ary tree can be implemented with left-child/right-sibling approach and this allows us to represent a k-ary tree with a binary tree ☺
  - Don't need to know about the number of children a node has
  - Easy to code as we have two children at most

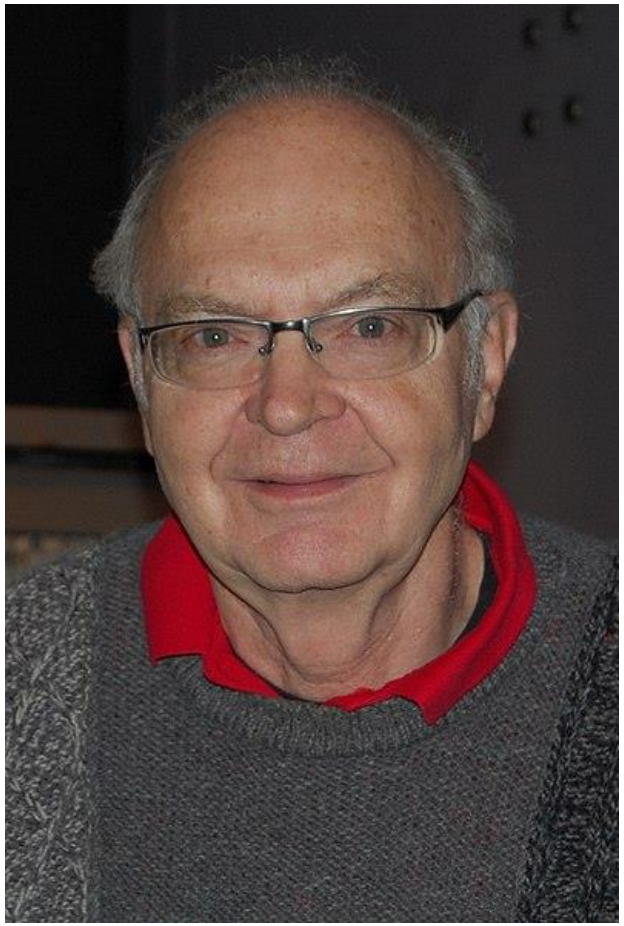


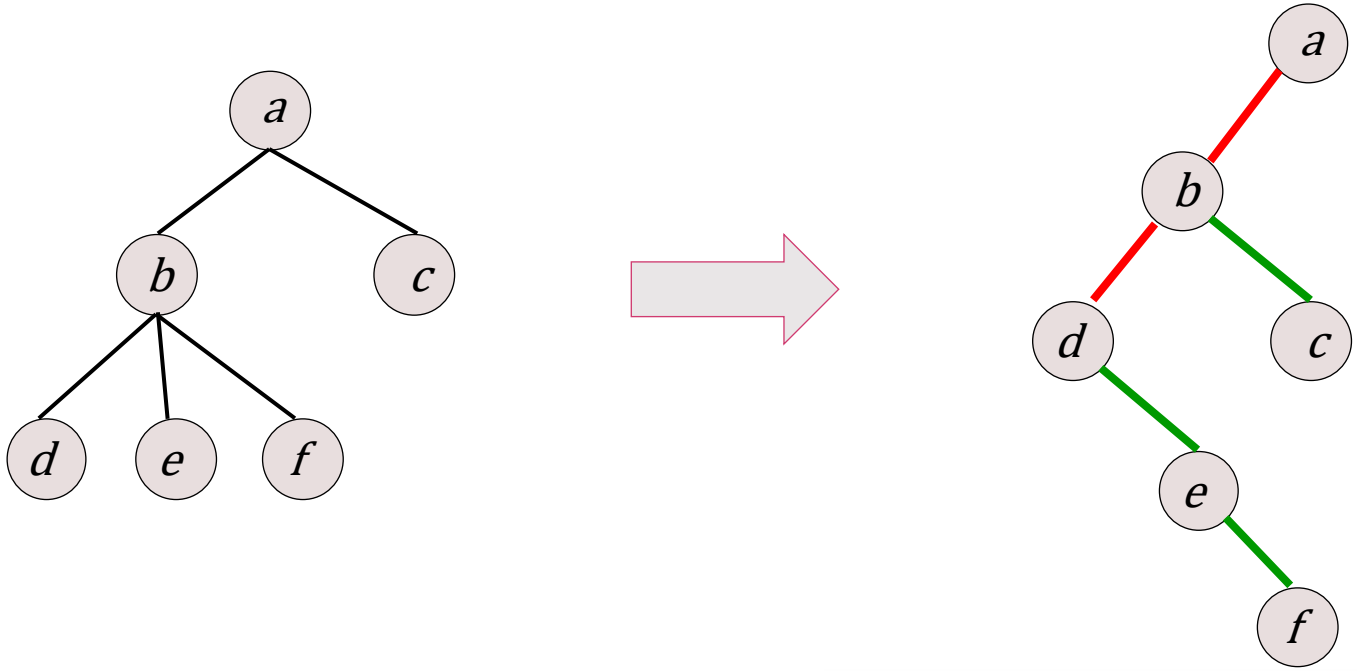
Left-Child/Right-Sibling Approach

- Step-1: Convert a (forest of) general tree into a binary tree
  - *Leftmost child* → Left child
  - *Right sibling* → Right child (tree roots are assumed to be siblings)
  - Rotate the *left-child/right-sibling* tree clockwise by 45 degrees
- So-called, Knuth transform

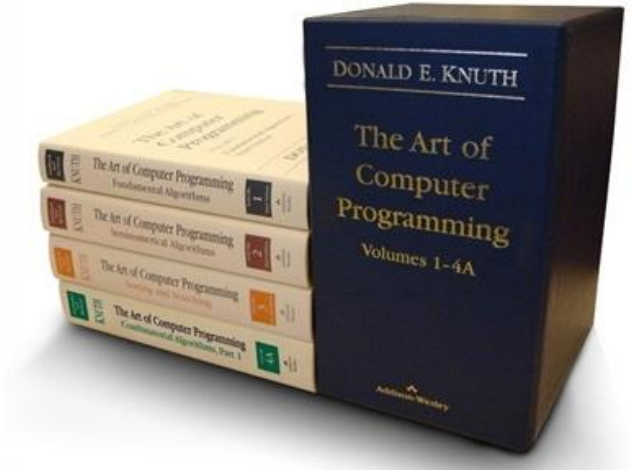
- Step-2: Linked implementation of the converted binary tree

Donald Knuth



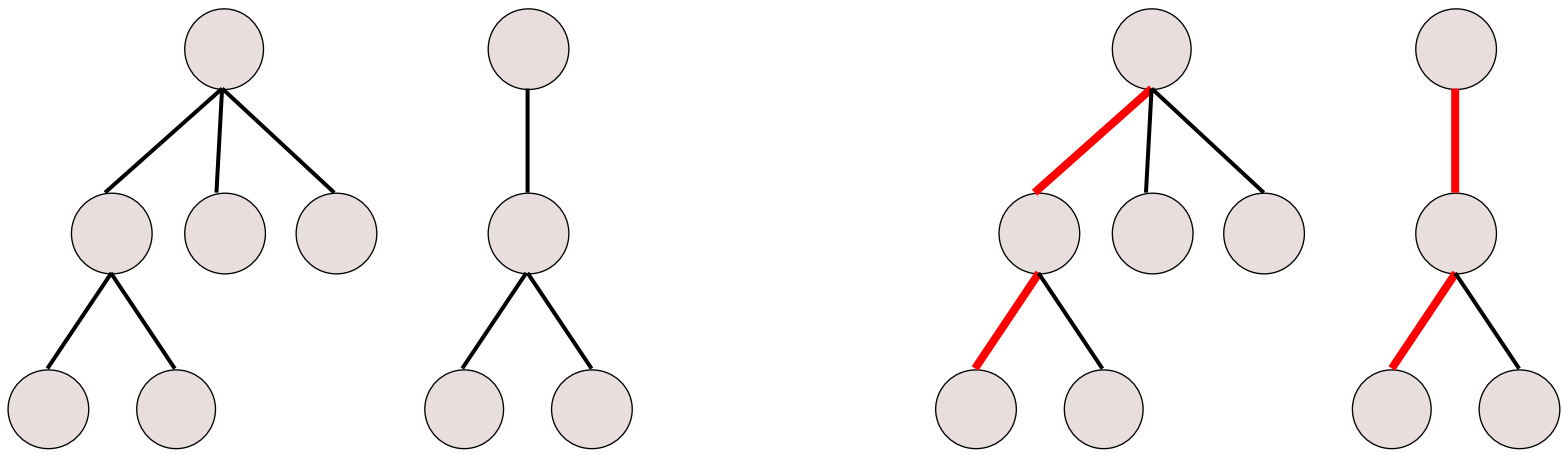


TEX



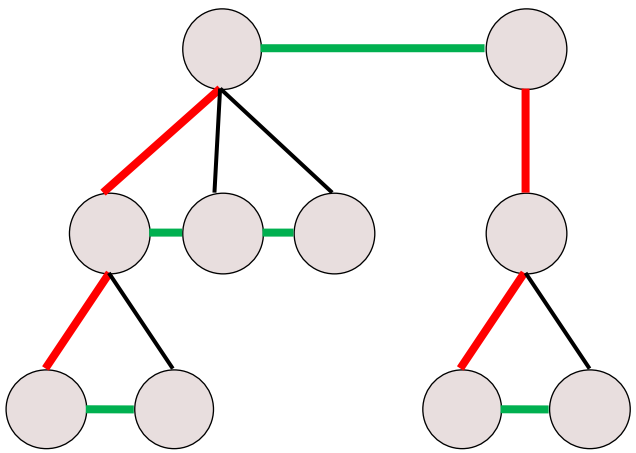
Converting into a Binary Tree (1)

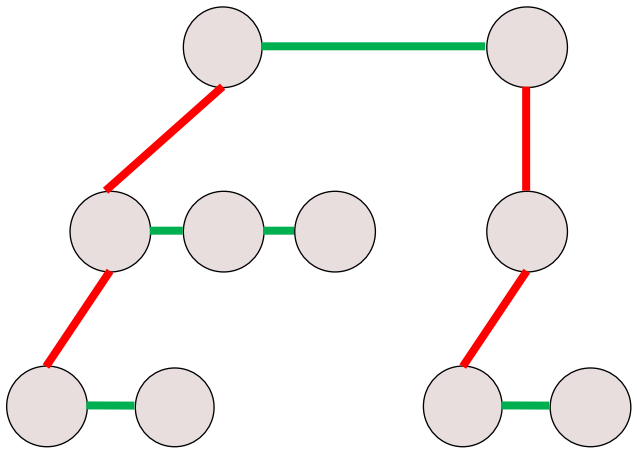
- Input: general trees
- *Leftmost child* → Left child



Converting into a Binary Tree (2)

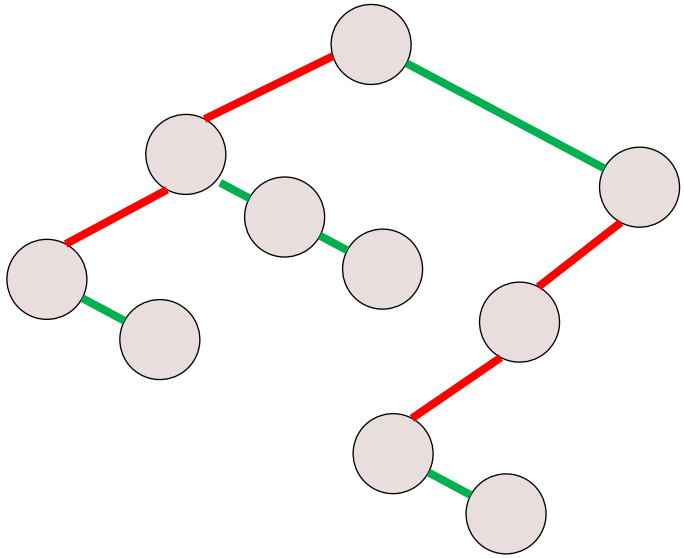
- *Right sibling* → Right child
  - Tree roots are assumed to be siblings
- Remove the other links



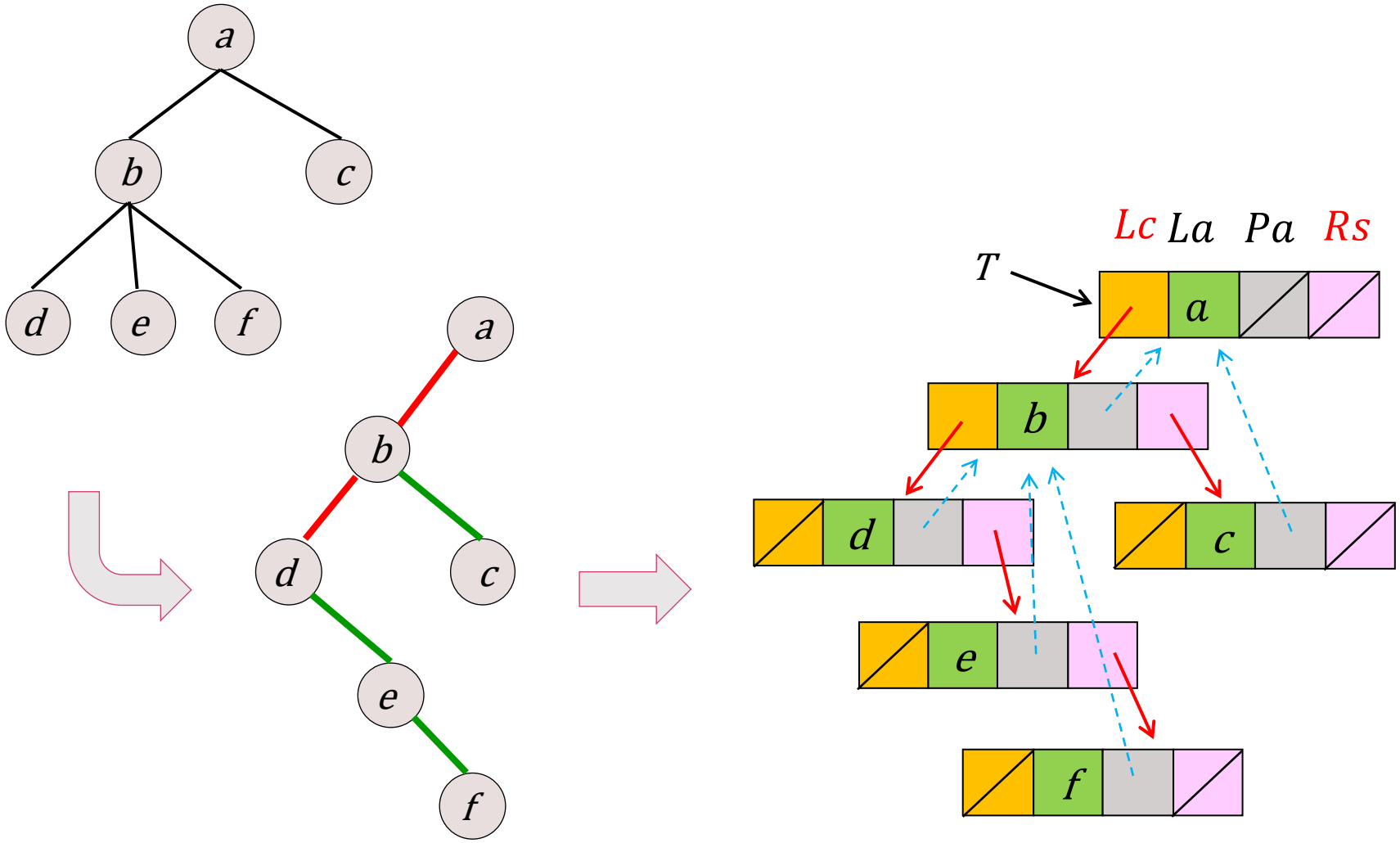


Converting into a Binary Tree (3)

- Rotate clockwise by $45°$
- Now ready for its linked implementation

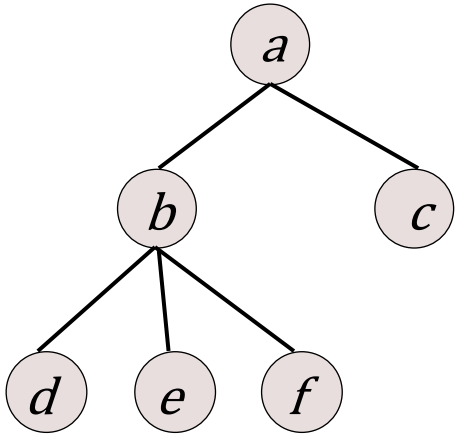


Left-Child/Right-Sibling Tree: Linked Rep.

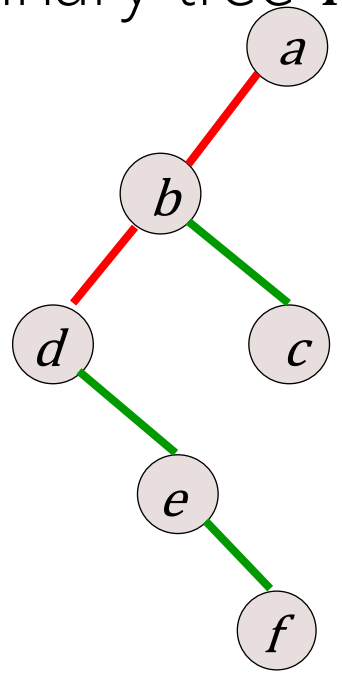


Traversal: Original vs. Converted Binary

- Original general tree T



- Converted binary tree T'

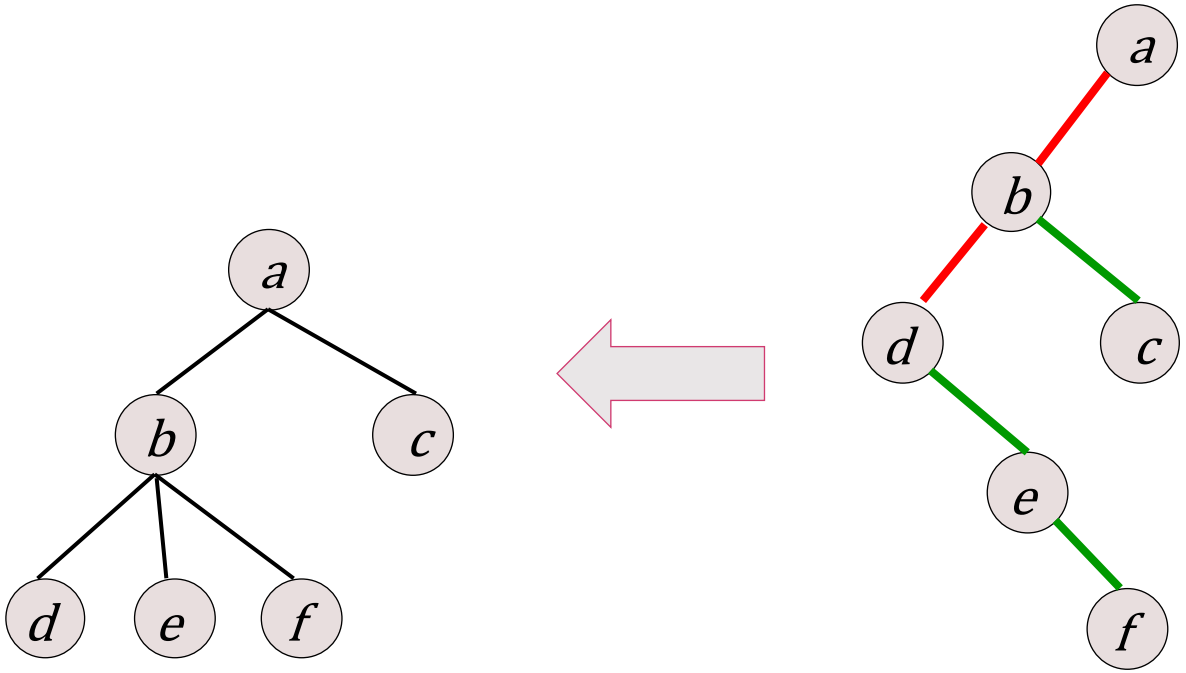


- $Preorder = a\ b\ d\ e\ f\ c$
- $Inorder = d\ b\ e\ f\ a\ c$
  $d\ e\ b\ f\ a\ c$
- $Postorder = d\ e\ f\ b\ c\ a$

- $Preorder = a\ b\ d\ e\ f\ c$
- $Inorder = d\ e\ f\ b\ c\ a$
- $Postorder = f\ e\ d\ c\ b\ a$

Quiz

- How to convert a converted binary tree to a general tree?



Summary

- Tree traversals
- Binary tree reconstruction
- General k-ary tree
- List-of-Children representation
- Left-child / right-sibling representation
- Time complexity of tree operations
- Knuth transform
- Static & dynamic tree implementations